STATISTICS AND PROBABILITY

ST331
(Statistical Computing 3)

ELEMENTARY STATISTICS

- A numeric variable is one whose observations are naturally recorded as numbers. There are two types of numeric variables: continuous and discrete
- Categorical variables are those that provide groupings that may have no logical order, or a logical order with inconsistent differences between groups

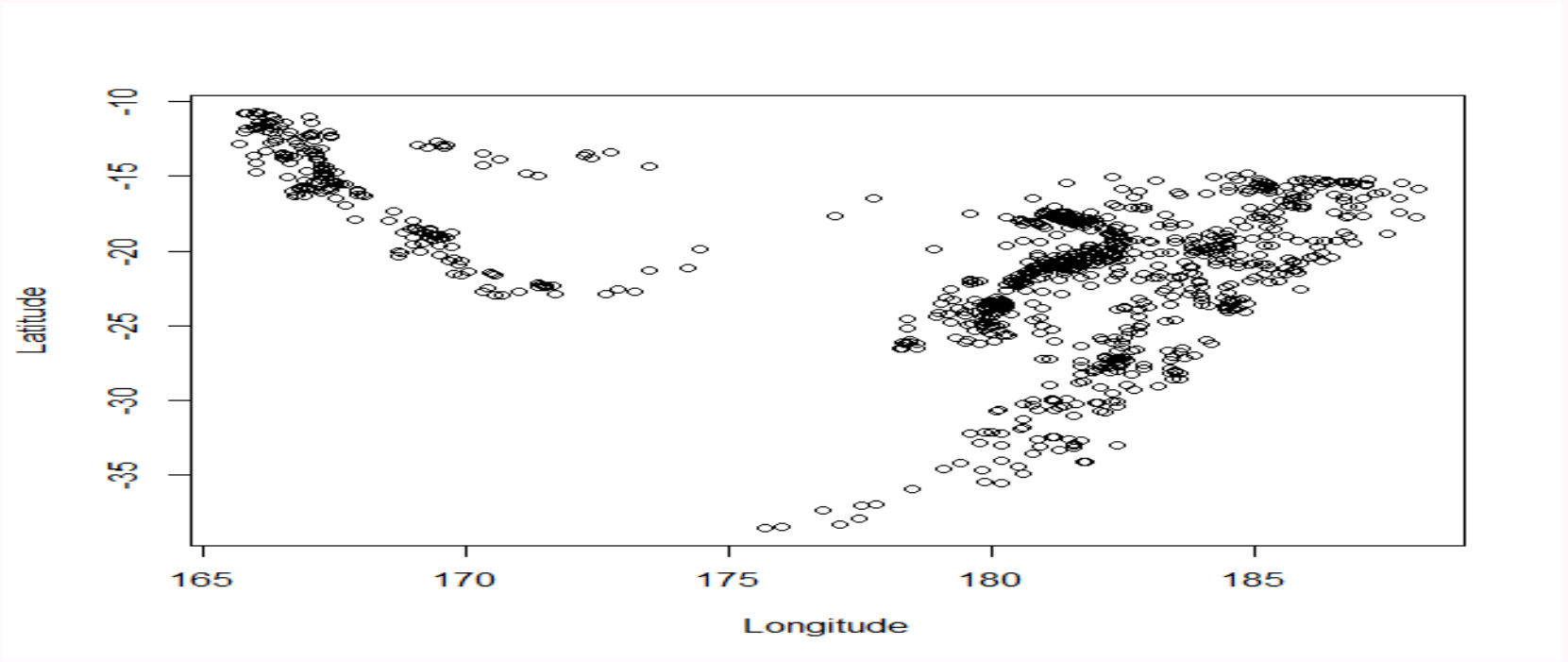
ELEMENTARY STATISTICS

- weight is a is a numeric-continuous variable while feed is clearly a categorical variable because it has only six possible outcomes, which aren't numeric.

```
> chickwts[1:5,]
  weight      feed
1    179 horsebean
2    160 horsebean
3    136 horsebean
4    227 horsebean
5    217 horsebean
> chickwts$weight
 [1] 179 160 136 227 217 168 108 124 143 140 309 229 181 141 260 203 148
[18] 169 213 257 244 271 243 230 248 327 329 250 193 271 316 267 199 171
[35] 158 248 423 340 392 339 341 226 320 295 334 322 297 318 325 257 303
[52] 315 380 153 263 242 206 344 258 368 390 379 260 404 318 352 359 216
[69] 222 283 332
> chickwts$feed
 [1] horsebean horsebean horsebean horsebean horsebean horsebean
 [7] horsebean horsebean horsebean horsebean linseed   linseed
[13] linseed   linseed   linseed   linseed   linseed   linseed
[19] linseed   linseed   linseed   linseed   soybean   soybean
[25] soybean   soybean   soybean   soybean   soybean   soybean
[31] soybean   soybean   soybean   soybean   soybean   soybean
[37] sunflower sunflower sunflower sunflower sunflower sunflower
[43] sunflower sunflower sunflower sunflower sunflower sunflower
[49] meatmeal  meatmeal  meatmeal  meatmeal  meatmeal  meatmeal
[55] meatmeal  meatmeal  meatmeal  meatmeal  meatmeal  casein
[61] casein    casein    casein    casein    casein    casein
[67] casein    casein    casein    casein    casein
Levels: casein horsebean linseed meatmeal soybean sunflower
```

ELEMENTARY STATISTICS

```
> quakes[1:5,]
     lat   long depth mag stations
1 -20.42 181.62   562 4.8       41
2 -20.62 181.03   650 4.2       15
3 -26.00 184.10    42 5.4       43
4 -17.97 181.66   626 4.1       19
5 -20.42 181.96   649 4.0       11
> plot(quakes$long,quakes$lat,xlab="Longitude",ylab="Latitude")
```



CENTRALITY: MEAN, MEDIAN, MODE

```
> xdata <- c(2,4.4,3,3,2,2.2,2,4)
> x.bar <- mean(xdata)
> x.bar
[1] 2.825
> m.bar <- median(xdata)
> m.bar
[1] 2.6
> xtab <- table(xdata)
> xtab
xdata
  2 2.2   3   4 4.4
  3   1   2   1   1
> min(xdata)
[1] 2
> max(xdata)
[1] 4.4
> d.bar <- xtab[xtab==max(xtab)]
> d.bar
2
3
```

Here, 2 is the value and 3 is the frequency of that value.

CENTRALITY: MEAN, MEDIAN, MODE

```
> mean(quakes$mag)
[1] 4.6204
> median(quakes$mag)
[1] 4.6
> Qtab <- table(quakes$mag)
> Qtab

  4 4.1 4.2 4.3 4.4 4.5 4.6 4.7 4.8 4.9   5 5.1 5.2 5.3 5.4 5.5 5.6 5.7 5.9   6 6.1 6.4
 46  55  90  85 101 107 101  98  65  54  47  43  29  21  20  14   9   8   2   3   1   1
> Qtab[Qtab==max(Qtab)]
4.5
107
```

COUNTS, PERCENTAGES, AND PROPORTIONS

```
> table(chickwts$feed)

    casein horsebean   linseed  meatmeal    soybean sunflower
        12        10        12        11         14        12
> table(chickwts$feed)/nrow(chickwts)

    casein horsebean   linseed  meatmeal    soybean sunflower
0.1690141 0.1408451 0.1690141 0.1549296 0.1971831 0.1690141
> table(chickwts$feed)

    casein horsebean   linseed  meatmeal    soybean sunflower
        12        10        12        11         14        12
> table(chickwts$feed)/nrow(chickwts)*100

    casein horsebean   linseed  meatmeal    soybean sunflower
 16.90141  14.08451  16.90141  15.49296  19.71831  16.90141
> table(chickwts$feed)/nrow(chickwts)

    casein horsebean   linseed  meatmeal    soybean sunflower
0.1690141 0.1408451 0.1690141 0.1549296 0.1971831 0.1690141
> round(table(chickwts$feed)/nrow(chickwts),digits=3)

    casein horsebean   linseed  meatmeal    soybean sunflower
     0.169     0.141     0.169     0.155     0.197     0.169
```

QUANTILES, PERCENTILES, AND THE FIVE NUMBER SUMMARY

```
> quantile(xdata,prob=0.8)
80%
3.6
> quantile(xdata,prob=c(0,0.25,0.5,0.75,1))
  0%  25%  50%  75% 100%
2.00 2.00 2.60 3.25 4.40
```

Here, you've used quantile to obtain what's called the five-number summary of xdata, comprised of the 0th percentile (the minimum), the 25th percentile, the 50th percentile, the 75th percentile, and the 100th percentile(the maximum).

```
> summary(xdata)
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
  2.000   2.000   2.600   2.825   3.250   4.400
> quantile(chickwts$weight,prob=c(0.25,0.75))
   25%    75%
204.5  323.5
```

This indicates that 25 percent of the weights lie at or below 204.5 grams and that 75 percent of the weights lie at or below 323.5 grams

COVARIANCE AND CORRELATION

Correlation measures the extent (strength) and direction of the relationship, while covariance only shows the direction, not the extent

Covariance

$$r_{xy} = \frac{1}{n-1}\sum_{i-1}^{n}(x_i - \bar{x})(y_i - \bar{y})$$

Correlation

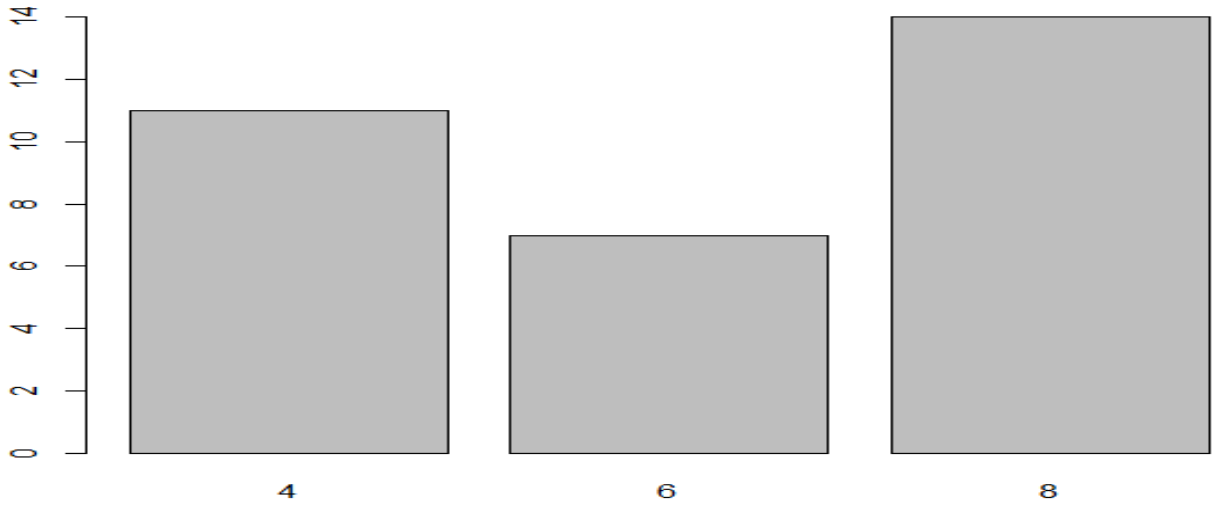
$$\rho_{xy} = \frac{r_{xy}}{s_x s_y},$$

```
> cov(xdata,ydata)
[1] 1.479286
> cor(xdata,ydata)
[1] 0.7713962
```

BASIC DATA VISUALIZATION (BARPLOTS AND PIE CHARTS)

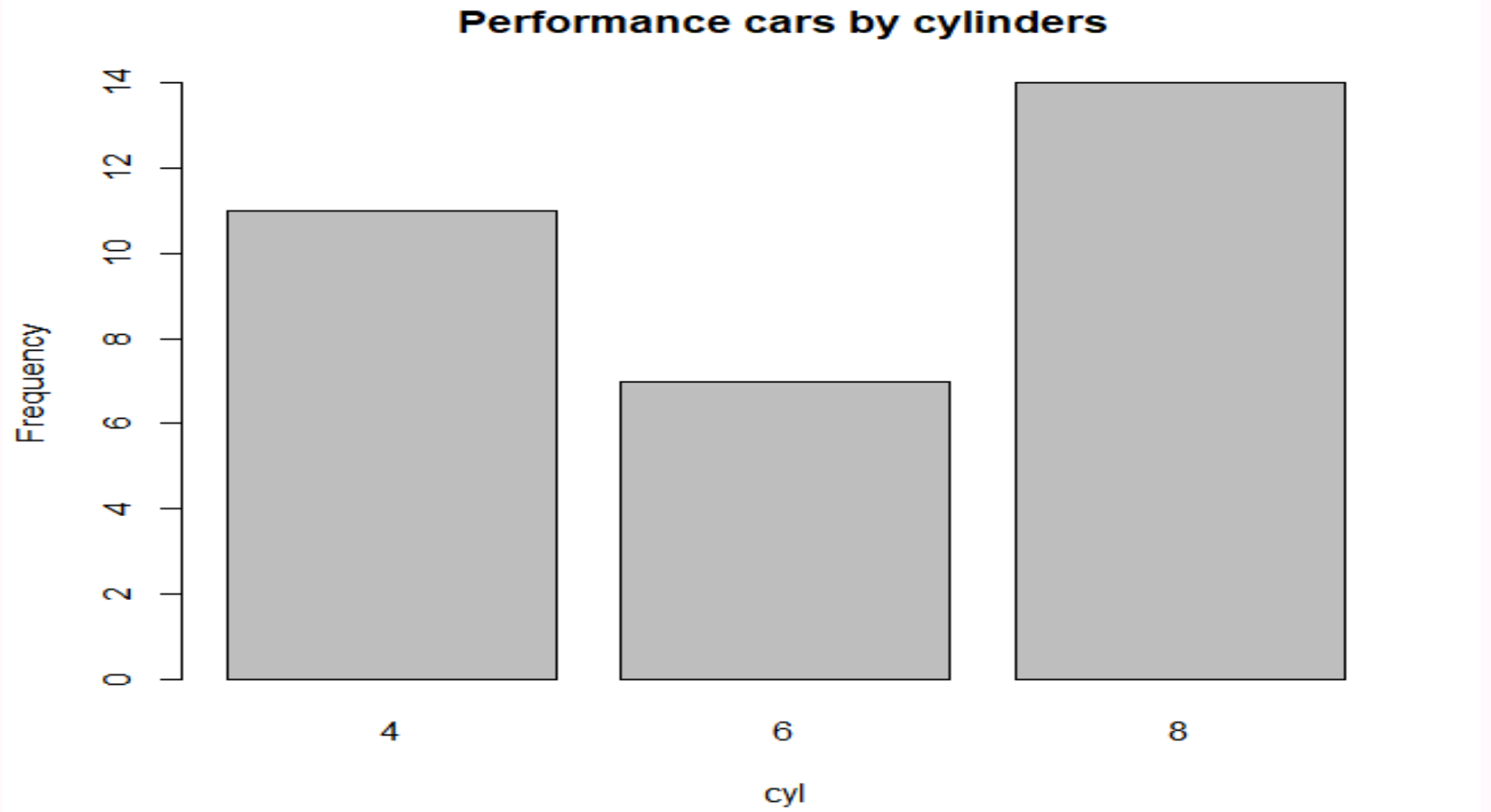
```
> mtcars[1:5,]
                   mpg cyl disp  hp drat    wt  qsec vs am gear carb
Mazda RX4         21.0   6  160 110 3.90 2.620 16.46  0  1    4    4
Mazda RX4 Wag     21.0   6  160 110 3.90 2.875 17.02  0  1    4    4
Datsun 710        22.8   4  108  93 3.85 2.320 18.61  1  1    4    1
Hornet 4 Drive    21.4   6  258 110 3.08 3.215 19.44  1  0    3    1
Hornet Sportabout 18.7   8  360 175 3.15 3.440 17.02  0  0    3    2
> cyl.freq <- table(mtcars$cyl)
> cyl.freq

 4  6  8
11  7 14
> barplot(cyl.freq)
```



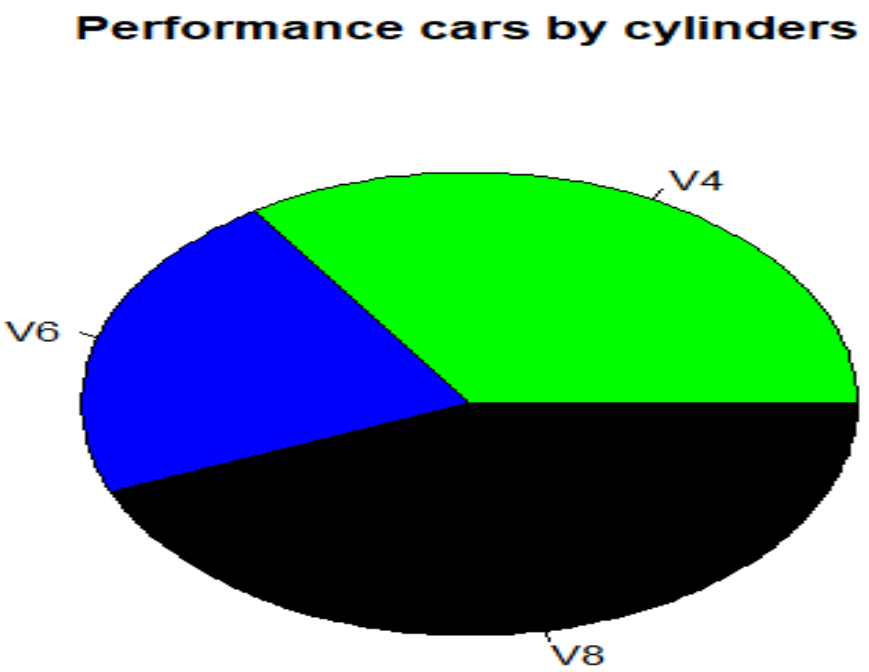
BASIC DATA VISUALIZATION (BARPLOTS AND PIE CHARTS)

```
> barplot(cyl.freq, xlab="cyl", ylab="Frequency", main = "Performance cars by cylinders")
```



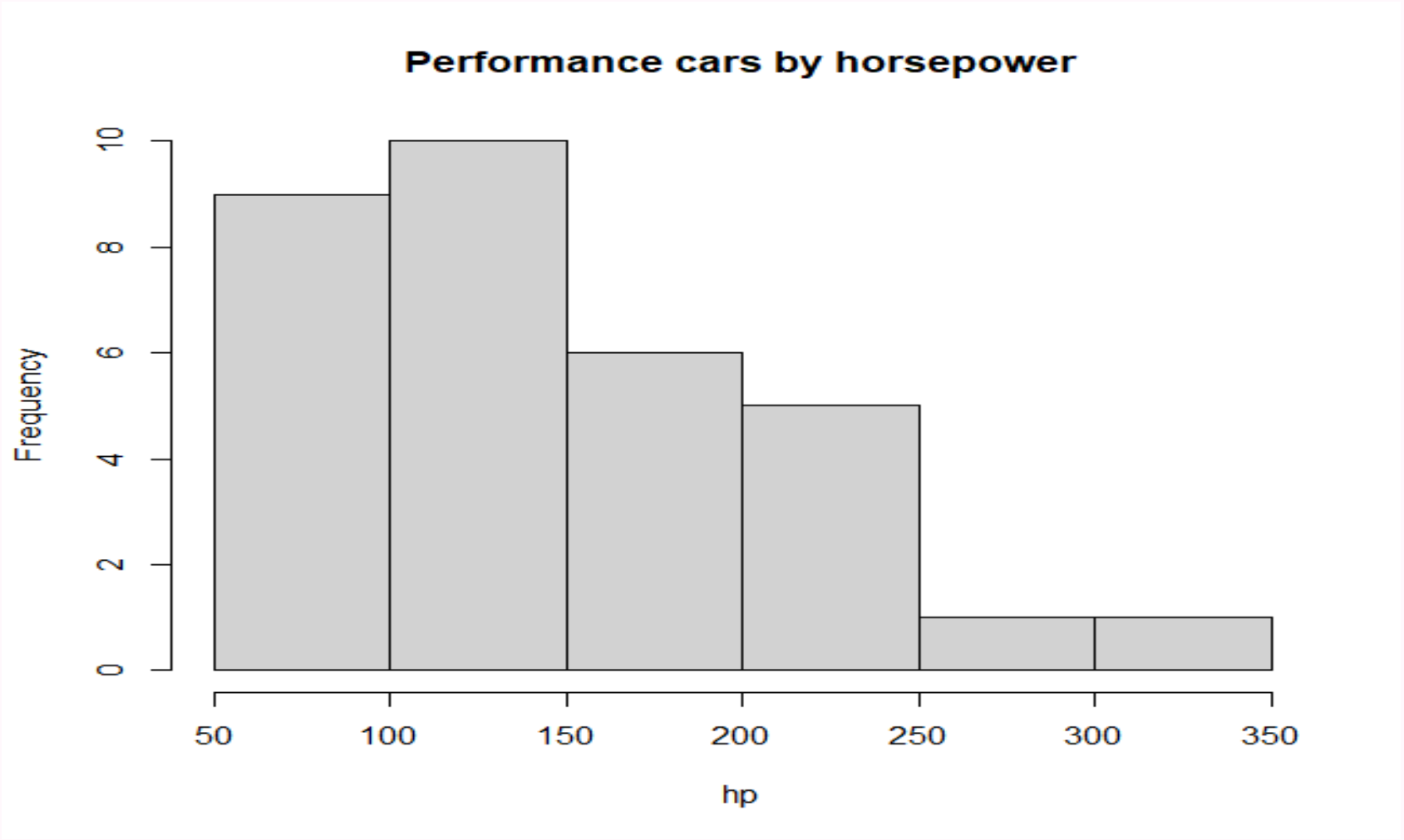
BASIC DATA VISUALIZATION (BARPLOTS AND PIE CHARTS)

```
> pie(table(mtcars$cyl),labels=c("V4","V6","V8"),
+     col=c("green","blue","black"),main="Performance cars by cylinders")
```



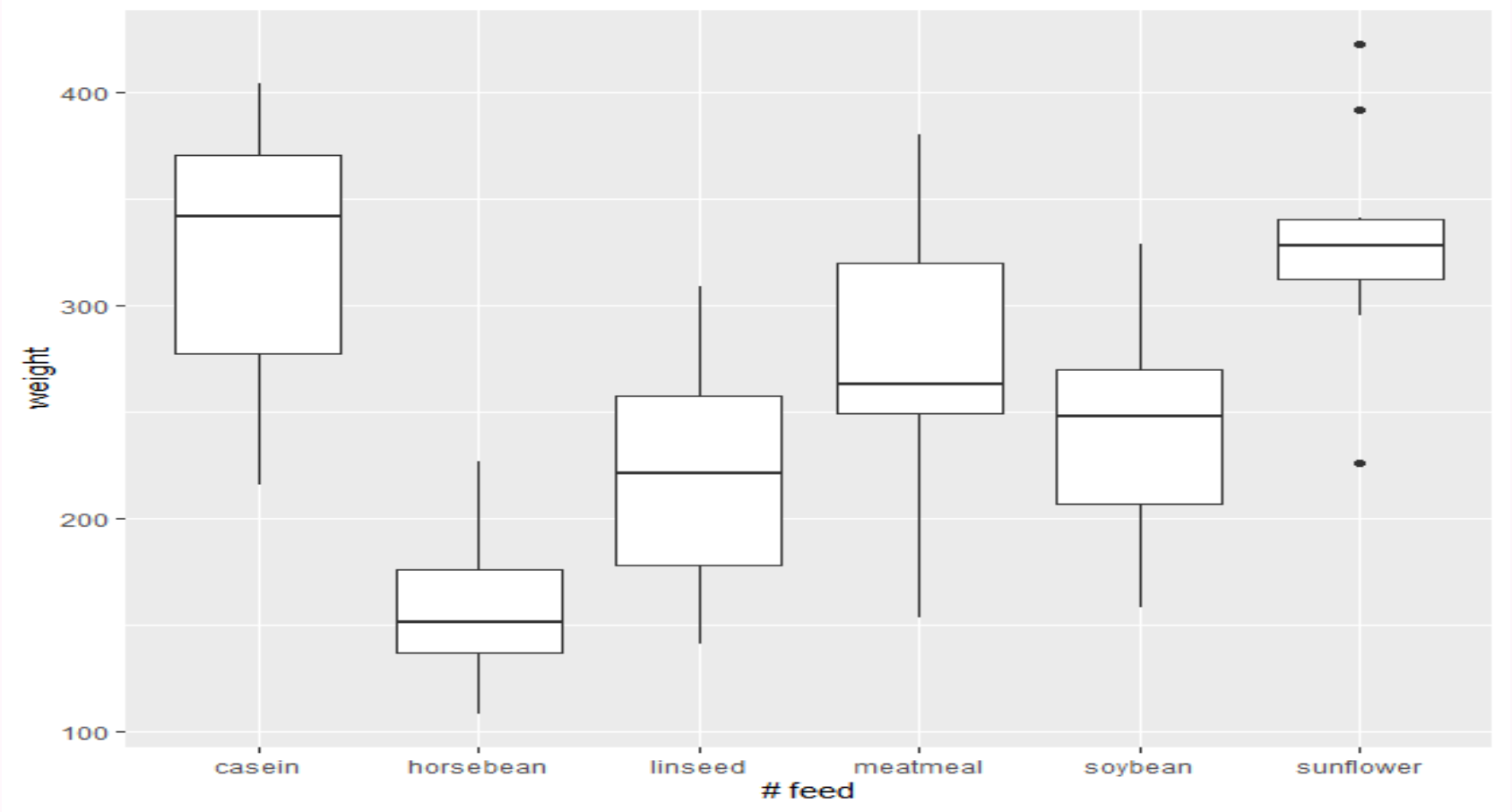
HISTOGRAM

```
> hist(mtcars$hp, xlab="hp", ylab="Frequency", main = "Performance cars by horsepower")
```



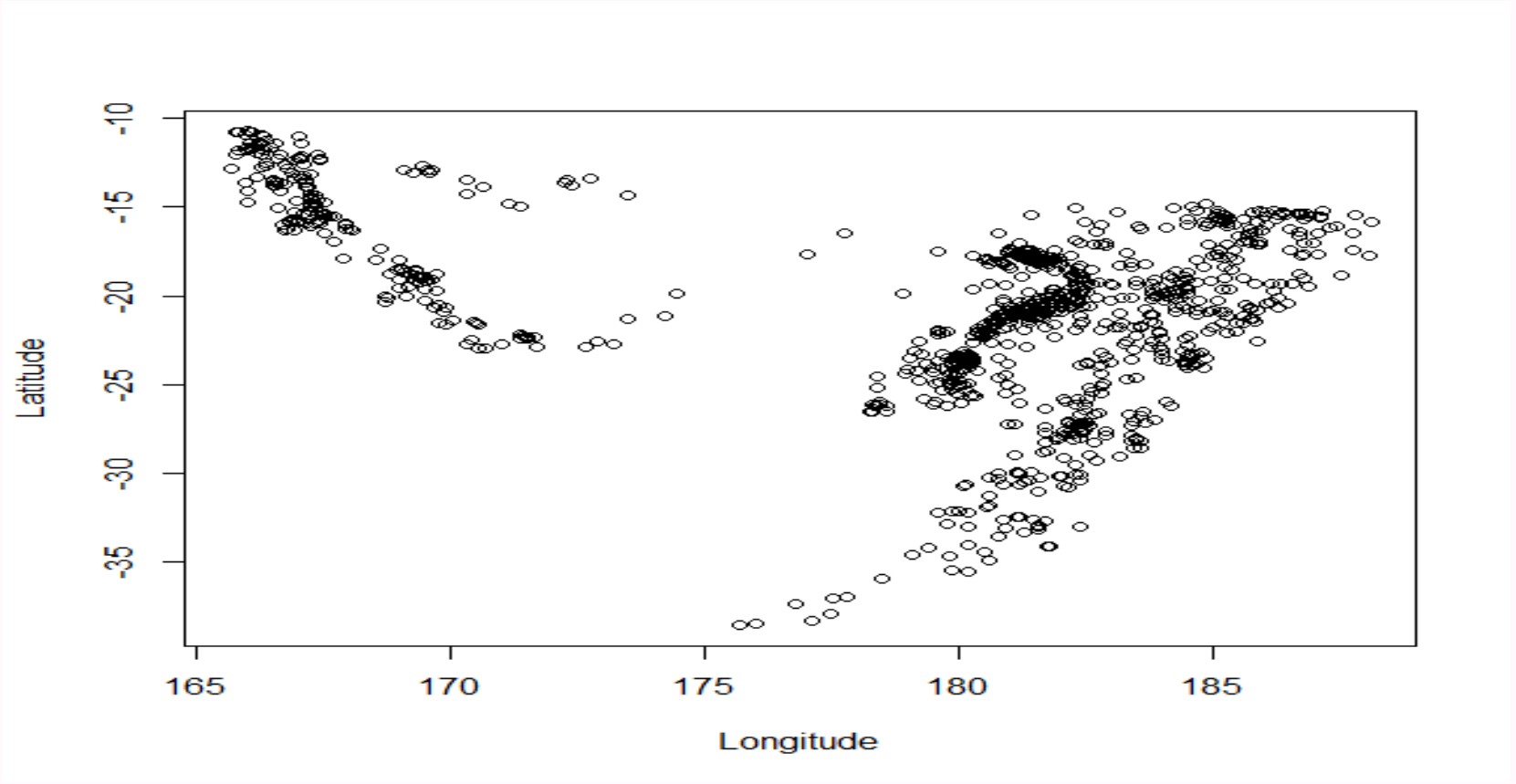
BOXPLOT

```
> library(ggplot2)
> qplot(chickwts$feed,chickwts$weight,geom="boxplot",xlab="# feed",ylab="weight")
```



SCATTERPLOTS

```
> plot(quakes$long,quakes$lat,xlab="Longitude",ylab="Latitude")
```



PROBABILITY

Intersection

When a dice is toss, what is the probability that on a single toss you roll an even number and it's a 4 or more?

```
> P_ANB= (2/3)*(1/2)
> P_ANB
[1] 0.3333333
```

Union

So, in rolling the die, what's the probability that you observe an even number or one that's at least 4

```
> P_AUB= (1/2)+(1/2)-(1/3)
> P_AUB
[1] 0.6666667
```

Complement

find the probability that you do not roll a 4 or greater

```
> P_A_= 1-1/2
> P_A_
[1] 0.5
```

CUMULATIVE PROBABILITY DISTRIBUTIONS OF DISCRETE RANDOM VARIABLES

A cumulative probability for a random variable X is "the probability of observing less than or equal to x" and written as $Pr(X \leq x)$

x	−4	0	1	8
$Pr(X = x)$	0.32	0.48	0.15	0.05
$Pr(X \leq x)$	0.32	0.80	0.95	1.00

```
> X.outcomes <- c(-4,0,1,8)
> X.prob <- c(0.32,0.48,0.15,0.05)
> X.cumul <- cumsum(X.prob)
> X.cumul
[1] 0.32 0.80 0.95 1.00
```

MEAN AND VARIANCE OF A DISCRETE RANDOM VARIABLE

Mean

$$\mu_X = \mathbb{E}[X] = x_1 \times \Pr(X = x_1) + \ldots + x_k \times \Pr(X = x_k)$$
$$= \sum_{i=1}^{k} x_i \Pr(X = x_i)$$

```
> mu.X <- sum(X.outcomes*X.prob)
> mu.X
[1] -0.73
```

Variance

$$\sigma_X^2 = \mathrm{Var}[X] = (x_1 - \mu_X)^2 \times \Pr(X = x_1)$$
$$+ \ldots + (x_k - \mu_X)^2 \times \Pr(X = x_k)$$
$$= \sum_{i=1}^{k} (x_i - \mu_X) \Pr(X = x_i)$$

```
> var.X <- sum((X.outcomes-mu.X)^2*X.prob)
> var.X
[1] 7.9371
> sd.X <- sqrt(var.X)
> sd.X
[1] 2.817286
```

CLASS EXERCISE

Suppose you construct the following table providing probabilities associated with the random variable S, the total stars given to any movie in a particular genre by one critic:

s	1	2	3	4	5
$P(S=s)$	0.10	0.13	0.21	?	0.15

i. Assuming this table describes the complete set of outcomes, evaluate the missing probability P(S=4)
ii. Obtain the cumulative probabilities.

iii. What is the mean of S, the expected number of stars this critic will award any given movie in this genre?

iv. What is the standard deviation of S?

v. What is the probability that any given movie in this genre will be given at least three stars?

ST431
(Design and Analysis of Experiment)

CLASS EXERCISE

```
> p <- c(0.10, 0.13, 0.21, NA, 0.15)
> missingvalue <- 1 - sum(p, na.rm = TRUE)
> missingvalue
[1] 0.41
> p <- c(0.10, 0.13, 0.21, 0.41, 0.15)
> s<-c(1:5)
> mean_S <- sum(s * p)
> mean_S
[1] 3.38
> var_S <- sum(s^2 * p) - mean_S^2
> var_S
[1] 1.3956
> sd_S <- sqrt(var_S)
> sd_S
[1] 1.181355
> sum(p[s >= 3])
[1] 0.77
```

ST431

(Design and Analysis of Experiment)